



Homework 6

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Decision Trees



Motivation

You already know what the goal of Classification is: in general, classification is the process in which ideas and objects are recognized, differentiated, and understood. In machine learning, classification is the problem of identifying to which of a set of categories (sub-populations) a new observation belongs, on the basis of a training set of data containing observations (or instances) whose category membership is known. An example would be assigning a given email into "spam" or "non-spam" classes or assigning a diagnosis to a given patient as described by observed characteristics of the patient (gender, blood pressure, presence or absence of certain symptoms, etc.) [https://en.wikipedia.org/wiki/Statistical_classification].

You also already know about several algorithms for classification and Regression, like naïve Bayes.

In this chapter, we will be covering another algorithms Decision Trees. We will see how to construct them and how to use them for Classification and Regression.

You will be studying and familiarizing yourself with the basics on your own in this homework, and we will be working together on the advanced material during the lecture.



Decision Trees for Classification

Estimated time: 20min to watch the video + 15 min for the questions

♦ **Watch the video ["Decision and Classification Trees, Clearly Explained!!!"](https://www.youtube.com/watch?v=_L39rN6gz7Y)**

https://www.youtube.com/watch?v=_L39rN6gz7Y

♦ **Please answer the following questions**

- 1) What is a decision tree, and how is it used in machine learning?
- 2) Can you explain the difference between classification trees and regression trees?
- 3) What are the key components (nodes, branches, leaves) of a decision tree, and what does each represent?
- 4) How does the process of splitting work in decision trees, and what criteria are commonly used to determine the best split?
- 5) What is the concept of 'impurity' in decision trees, and how do metrics like Gini impurity and entropy relate to it?
- 6) How can overfitting occur in decision trees, and what techniques are used to prevent it?
- 7) How do decision trees handle continuous (numerical) vs. categorical data?

1) a decision tree is a way to classify and predict the class (category) or numerical value of a data set by evaluating its data via statements which can be either true or false and making a decision based on that

2) classification trees are used to classify things into categories whereas regression trees predict numerical values

3) the statements within a tree are the nodes and can either be the root node (start of the tree), an internal node (also called branch) which are the decision points between the root and the leaf nodes (have arrows leading in and out) and leaves which are the categories and result of the tree

4) splitting is the concept of dividing a data set into smaller subsets based on metrics like Gini Impurity to better predict the class of the data, goal is to create pure as can be subsets that predict the given class accurately

5) impurity refers to the concept of a node not uniformly classifying data as belonging to a class or not (based on the training set), i.e. it gives no pure true/false classification and is instead mixed, gini impurity and entropy are metrics with which to evaluate the impurity of the node

6) overfitting can occur when the subset a node is based on while not being very impure is very small and does not guarantee a good classification, can be counteracted by requiring the node to represent a certain threshold of people

7) for categorical data which are either true or false for the data, they are taken as is

on the other hand, for numerical data a threshold is chosen and the data set is evaluated based on that threshold (cf. ROC chart) the best threshold is chosen via a metric such as gini impurity



Decision Trees for Regression

Estimated time: 25min to watch the video + 15 min for the questions

♦ **Watch the video "Regression Trees, Clearly Explained!!!"**

<https://www.youtube.com/watch?v=g9c66TUyIZ4>

♦ **Please answer the following questions**

- 1) What is a regression tree, and how does it differ from a classification tree?
- 2) How can regression trees be used for predicting continuous outcomes?
- 3) How does a regression tree determine the best split at each node?
- 4) What are some common methods to prevent overfitting in regression trees?

- 1) used for continuous target values, has numerical output
- 2) splits data recursively into smaller more homogenous groups and outputs assigned predicted value based on thresholds
- 3) based on thresholds and sum of squared residuals
- 4) occurs to too granular splits that capture noise instead of overall trend, can be fixed by setting minimum sample size per node/ split



Entropy

You know the concept of information entropy as introduced by Claude Shannon: Shannon's theory defines a data communication system composed of three elements: a source of data, a communication channel, and a receiver. The "fundamental problem of communication" – as expressed by Shannon – is for the receiver to be able to identify what data was generated by the source, based on the signal it receives through the channel. The entropy is often described as the minimum number of bits you need to transmit the information.

Information entropy is also frequently used to determine the split in the construction of a decision tree, this criterion is called Information Gain. It is easier to understand when looking at Entropy as "expected surprise".

Estimated time: 20min to watch the video

- ♦ **Watch the video ["Entropy \(for data science\) Clearly Explained!!!"](https://www.youtube.com/watch?v=YtebGVx-Fxw)**
<https://www.youtube.com/watch?v=YtebGVx-Fxw>

Estimated time: 20min to read the Blog Post

- ♦ **Read the Blog Article ["A Simple Explanation of Information Gain and Entropy"](https://victorzhou.com/blog/information-gain/)**
<https://victorzhou.com/blog/information-gain/>



Congratulations!

You now understand the basic idea of Decision Trees, one of the most important Classification Algorithms. You know how to build them and how to use them.

In the lecture, we will be looking at making them even better (i.e. increasing their classification accuracy) by combining multiple Decision Trees in really smart ways...